

Program 8

Objective: Create Kubernetes manifest files and demonstrate deployment creation, service exposure, logs, and description of resources.

Project Structure

```
Pr8/  
├──  
└── nginx_dep.yml
```

Kubernetes Manifest File

Deployment and Service (nginx_dep.yml)

```
apiVersion: apps/v1  
kind: Deployment  
metadata:  
  name: nginx-deployment  
spec:  
  replicas: 2  
  selector:  
    matchLabels:  
      app: nginx  
  template:  
    metadata:  
      labels:  
        app: nginx  
    spec:  
      containers:  
        - name: nginx  
          image: nginx:latest  
          ports:  
            - containerPort: 80  
---  
apiVersion: v1  
kind: Service  
metadata:  
  name: nginx-service  
spec:
```

```
spec:
  type: NodePort
  selector:
    app: nginx
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30080
```

Steps and Execution

Step 1: Apply the Manifest File

```
1rv24mc078_prateek_kumar@arch kubectl apply -f nginx_dep.yml
deployment.apps/nginx-deployment created
service/nginx-service created
```

Step 2: Check Deployment Status

```
1rv24mc078_prateek_kumar@arch kubectl get deployments
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	2/2	2	2	40s

Step 3: View Pod Status

```
1rv24mc078_prateek_kumar@arch kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-6f96d4446b-a1c9d	1/1	Running	0	35s
nginx-deployment-6f96d4446b-b7x4m	1/1	Running	0	35s

Step 4: View Logs

```
1rv24mc078_prateek_kumar@arch kubectl logs nginx-deployment-6f96d4446b-a1c9d
/docker-entrypoint.sh: Configuration complete; ready for start up
```

Step 5: Describe Deployment

```
1rv24mc078_prateek_kumar@arch kubectl describe deployment nginx-deployment
```

```
Name:                nginx-deployment
Namespace:           default
Replicas:            2 desired | 2 updated | 2 total | 2 available
StrategyType:        RollingUpdate
Pod Template:
  Containers:
    nginx:
      Image:          nginx:latest
      Port:           80/TCP
Conditions:
  Available           True
  Progressing         True
```

Step 6: Access Application

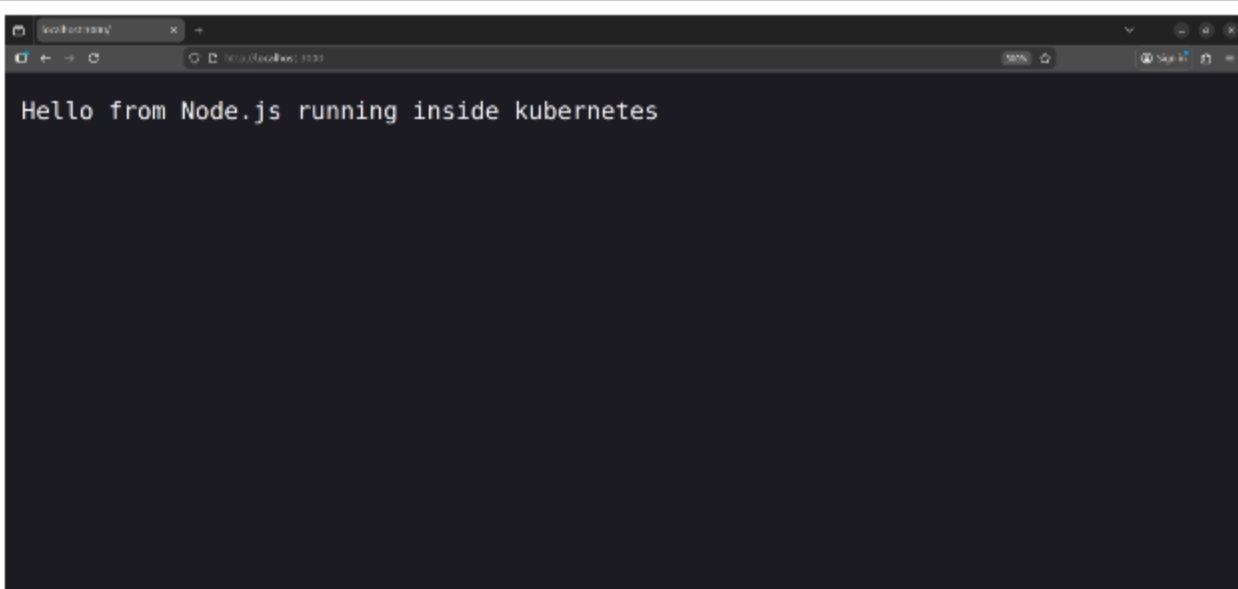
```
1rv24mc078_prateek_kumar@arch minikube service nginx-service
```

-----	-----	-----	-----
NAMESPACE	NAME	TARGET PORT	URL
-----	-----	-----	-----
default	nginx-service	80	http://192.168.49.2:30080
-----	-----	-----	-----

Opening service default/nginx-service in default browser...

Output

Nginx Application Output in Browser:



A screenshot of a web browser window. The address bar shows a URL. The main content area of the browser displays the text "Hello from Node.js running inside kubernetes" in a monospaced font on a dark background.



Result

The Kubernetes manifest file was successfully applied. The deployment, service, logs, and resource description were verified, and the Nginx application was accessed through the browser.